INTEGRATED APPLICATION · FEATURE DELIVERY · NOVEMBER 2026

Flood Risk & Disaster Management

Progress report | Implemented features, datasets used and models trained

PREPARED BY
Faiza Muzaffar

REPORTING PERIOD
November 2026

STATUS
Active development

REPORT SUMMARY

One integrated system covers flood monitoring, GIS risk screening, early warning, citizen reporting and incident response, data stewardship, river-flow forecasting and flood simulation with exposure analysis. Page 1 lists the implemented features; page 2 lists the datasets used and the models trained, with their current evidence status.

Implemented app features

- **Operational dashboard —** KPI and status panels, live refresh, provider-error visibility, hourly analytics ticker, activity timeline and a guided tour for new users.
- **Weather & river monitoring —** Open-Meteo rainfall, temperature, humidity and soil moisture; GloFAS modelled discharge kept separate from measured FFD readings; station hydrographs.
- **Flood-risk screening —** rule-based station risk score with severity and contributing factors; national weather-based GIS screening of 126 Pakistan districts.
- **GIS operations & map tools —** district, river, glacier, catchment and rainfall layers; coordinate picker, catchment study mode and 3D terrain views of Tarbela, Mangla, Marala and Taunsa.
- **Priority monitoring network —** 28 sites (15 barrages, 6 upstream stations, 3 dams, 4 confluences) with hover cards, 48-hour trends, 3 h / 24 h change and stale-reading flags.
- **Forecast & outlook views —** 5-day station discharge forecast window with CSV export, Upstream Source Flood Watch with flood-threshold lines, monthly Marala outlook and a 10-year risk outlook.
- **Citizen reports, alerts & incidents —** reports saved to the database with map location and photo-geotag extraction; mark important or close; alert approval before escalation; incident tracking.
- **Satellite water-extent analysis —** Sentinel-1 VV/VH with elevation, slope and permanent-water inputs produces a U-Net water overlay with area summary for any selected location.
- **Scenario preparation & simulation —** scenario editor with JSON import/export, checksum-tracked input registry, domain preparation and engine binding for HEC-RAS 2D, LISFLOOD-FP, HEC-HMS and ANUGA.
- **Model-result review —** map playback of depth, water-surface and speed rasters with hydrograph, 1x/4x/16x speed and A/B comparison of runs.
- **Flood visualisation & exposure —** storm replay, flood-spread and affected-population views, cinematic 3D; Jhang exposure study of population, roads, buildings and facilities.
- **Data observatory —** source, checksum and quality flags for every dataset; linked discharge and rainfall charts and a daily data-availability calendar from 1970 to 2026.
- **Platform —** FastAPI backend with 55+ endpoints, SQLite storage (PostgreSQL-ready), React map front ends and 54 automated test suites.

DATASETS · FORECASTING MODELS · EVIDENCE STATUS

Datasets used & models trained

Every dataset keeps its source, licence and checksum. Questionable records are withheld, never altered, and modelled values are never mixed into observed tables.

Datasets used

DATASET	SOURCE	PERIOD / SIZE	USED FOR
Marala daily inflow	WAPDA, cross-checked with IRSA	Mar 2016 – Dec 2024; 3,216 dates, 3,209 eligible	Forecast target
Marala 06:00 gauge	FFD / PMD daily bulletins	Jul 2024 – Oct 2026; 322 readings	Station forecast
Catchment rainfall	CHIRPS v3 (primary); ERA5-Land, IMERG, PERSIANN-CDR, NASA POWER	1990 – 2026, daily	Model input
Weather & snow	ERA5-Land temperature, dewpoint, radiation, PET, snow; MODIS snow cover	1980 – 2024, daily	Model input
Flood events	Global Flood Database, DesInventar, PMD peak records	GFD: 28 events (2003–2018); DesInventar: 10,097 rows	Risk models
Pretraining basins	Caravan (global), CAMELS-US	Multi-basin, daily	Transfer learning
Satellite imagery	Sentinel-1 RTC, Copernicus DEM, JRC surface water; Sen1Floods11	On demand; labelled chips	Water mapping
Terrain & rivers	FABDEM, WorldCover, HydroRIVERS, HydroBASINS, SoilGrids, 17 FFC reports	Kabul / Nowshera	Simulation
Exposure	WorldPop 2025, OpenStreetMap, Microsoft buildings, OCHA boundaries, UNOSAT	Jhang district	Impact estimates
Live feeds	Open-Meteo weather, GloFAS discharge	National, refreshed	Dashboard, screening

Models trained

- **River-flow forecasting (Marala) —** 36 configurations over 10, 15 and 30-day horizons: weather-only, past-discharge, encoder-decoder, elevation-band and pretrained (Caravan, CAMELS-US) LSTMs; HBV / delta-HBV hybrids; N-HiTS; N-BEATSx; gradient boosting; persistence and climatology baselines.
- **Training set-up —** 8 daily inputs with 365-day history; train 2016–2020, validate 2021, test 2022–2024; ERA5-Land with added noise stands in for future weather.
- **Key results —** day-10 flood hit rate 66.7% (catchment-mean LSTM) against 27.5% for persistence; best lead-10 NSE 0.876 (fine-tuned Caravan LSTM); day-30 flood recall 74.5% (HBV-LSTM residual).
- **Marala station & seasonal —** ridge/XGBoost on 322 gauge readings: 9.5% lower RMSE than persistence at 7 days; 365-day seasonal outlook trained on the GloFAS reach series.
- **Flood-risk XGBoost —** synthetic development model (6,000 rows, disabled); real-data classifiers on 2,410 GFD rows, 10,097 DesInventar rows and a 4,597-row Lower Indus pilot. Validation gates failed, so the rule-based engine stays operational.
- **Station discharge XGBoost —** Trimmu +24 h and Balloki +48 h baseline from lagged flow, rainfall, upstream flow, releases and soil moisture; bootstrap data are synthetic.
- **SAR U-Net water mapping —** Sen1Floods11 test IoU 65.36% and Dice 79.05%; Pakistan test IoU 13.47%.
- **Hydrologic & hydraulic —** HEC-HMS routing to Nowshera: withheld NSE 0.678, KGE 0.462 (target not met). HEC-RAS 2D published 385 hourly frames; LISFLOOD-FP 100 m screening; ANUGA 10,000-cell Jhang trial.

Data priority: reviewed sub-daily gauge series, surveyed channel geometry, archived forecast weather and independent flood extents are needed to calibrate and validate the models for operational use. Research scores above are retrospective and are not live forecast accuracy.